



UNIT PURPOSE

Students learn how measurements become defensible evidence. The unit moves from tools, units, and notation to factor-label reasoning, density, percent error, and evidence-based claims about accuracy and precision.

INSTRUCTIONAL RHYTHM

Engage -> Teach -> Model -> Guided Practice -> Independent Practice -> Exit Ticket -> Mixed Review -> Unit Test

SUGGESTED PACING

- 1 Lesson 1: Measurement Evidence and SI Records. Use Slides 1-3, Student Pages 1-2, and the exit ticket.
- 2 Lesson 2: Scientific Notation and Significant Figures. Model rounding only after the full calculation.
- 3 Lesson 3: Dimensional Analysis. Require units on every factor and a visible cancellation check.
- 4 Lesson 4: Density and Percent Error. Connect calculations to measured evidence and units.
- 5 Lesson 5: Accuracy, Precision, and Error. Compare repeated trials before making a claim.
- 6 Cumulative mixed review, feedback, and targeted reteaching.
- 7 Unit Test A. Reserve Form B for a retake, alternate class, or test security.

PACKAGE MAP

Projectable slides teach the lesson. Student pages provide guided notes, modeled thinking, supported practice, independent practice, and exit tickets. Tests A and B follow all instruction and review. The teacher file contains every answer and rationale.



Support, monitor, and respond

BEFORE TEACHING

Decide calculator expectations and the significant-figure convention used by your course. Confirm students can multiply powers of ten. Preview the pictured tools and remind students that the correct tool depends on the quantity and needed resolution.

FORMATIVE LOOK-FORS

Students attach units to measured values, distinguish exact conversion factors from measured data, cancel units before calculating, round once at the end, and use repeated trials when discussing precision.

SUPPORT

Keep the prefix table, factor-label frame, and formula bank visible. Let students annotate units in color. Reduce arithmetic load without removing the reasoning step.

EXTEND

Ask students to design a repeated-trial measurement plan, predict the effect of a systematic offset, and defend which additional data would strengthen an accuracy claim.

ASSESSMENT USE

Use Form A after instruction and mixed review. Reserve Form B for retakes, an alternate class period, or test security. Do not use either test as the first exposure to a skill.

SCOPE

Supports the NGSS practice of Using Mathematics and Computational Thinking; no formal performance-expectation alignment is claimed.

L1-1. Classify 'The solution is pale blue' and 'The solution volume is 18.5 mL' as qualitative or quantitative.

ANSWER: 'Pale blue' is qualitative. '18.5 mL' is quantitative because it includes a measured value and unit.

L1-2. Choose the pictured tool best suited to measure 24.6 mL of water. Cite one visual feature.

ANSWER: The graduated cylinder; its narrow body and graduated scale support a more precise volume reading than a beaker.

L1-3. Repair each record: (a) .45g (b) 6.0ml (c) 5 kgs.

ANSWER: (a) 0.45 g (b) 6.0 mL (c) 5 kg.

L1-4. A student needs the mass of a dry solid. Explain why a graduated cylinder is not the correct first tool.

ANSWER: A graduated cylinder measures volume. An electronic balance directly measures mass.

L1-5. Write one complete quantitative record for a sample with a mass display of 7.250 g.

ANSWER: Example: sample mass = 7.250 g, measured with an electronic balance.

MISCONCEPTION

A number automatically makes an observation scientific and complete.

REPAIR

Require students to underline the value once and circle the unit; if either mark is missing, the record is incomplete.

L2-1. Write 0.0005070 in scientific notation and state its significant figures.

ANSWER: 5.070×10^{-4} ; four significant figures.

L2-2. State the number of significant figures in (a) 0.004060 (b) 700. (c) 1.0200 (d) 0.030.

ANSWER: (a) 4 (b) 3 (c) 5 (d) 2.

L2-3. Calculate $45.60 / 2.5$ and round correctly.

ANSWER: 18; the calculator result 18.24 is rounded to two significant figures.

L2-4. Calculate $18.42 + 3.7 + 0.006$ and round correctly.

ANSWER: 22.1; the tenths place limits the sum.

L2-5. A student rounds every intermediate step. State one risk and one repair.

ANSWER: Early rounding can accumulate error. Keep guard digits and round once after the final operation.

MISCONCEPTION

All zeros are either always significant or never significant.

REPAIR

Have students label each zero as leading, captive, or trailing before counting significant figures.

L3-1. Convert 3.45 km to meters using a visible conversion factor.

ANSWER: $3.45 \text{ km} \times (1000 \text{ m} / 1 \text{ km}) = 3.45 \times 10^3 \text{ m}$.

L3-2. Convert 0.0825 L to milliliters.

ANSWER: 82.5 mL.

L3-3. Convert $2.60 \times 10^5 \text{ mg}$ to grams and preserve significant figures.

ANSWER: $2.60 \times 10^2 \text{ g}$, or 260. g.

L3-4. Convert 18.0 m/s to km/h.

ANSWER: 64.8 km/h.

L3-5. Explain why a setup that ends in h/s is incomplete when the target is m/s.

ANSWER: The distance unit was not converted or canceled correctly; dimensional consistency requires only m/s to remain.

MISCONCEPTION

Conversion means moving a decimal according to a memorized direction.

REPAIR

Cover the numbers and inspect only the units. Reorient factors until every unwanted unit cancels.

Practice answers and repair moves

L4-1. A 31.6 g sample occupies 25.5 mL. Calculate density.

ANSWER: $31.6 \text{ g} / 25.5 \text{ mL} = 1.24 \text{ g/mL}$.

L4-2. A liquid has density 1.24 g/mL and volume 25.5 mL. Calculate mass.

ANSWER: $m = dV = 31.6 \text{ g}$.

L4-3. A solid has mass 14.9 g and density 2.50 g/cm³. Calculate volume.

ANSWER: $V = m/d = 5.96 \text{ cm}^3$.

L4-4. An experimental value is 12.4 g/mL and the accepted value is 12.0 g/mL. Calculate percent error.

ANSWER: $|12.4 - 12.0| / 12.0 \times 100 = 3.33\%$.

L4-5. Explain why percent error uses the accepted value in the denominator.

ANSWER: The accepted value provides the reference scale, so the difference is expressed relative to what is expected.

MISCONCEPTION

Density is the same as mass, or a larger sample must have a larger density.

REPAIR

Compare two amounts of the same material and calculate m/V for each; the ratio stays constant while mass and volume change.

L5-1. Trials are 10.01, 10.00, and 10.02 g; accepted mass is 10.01 g. Classify accuracy and precision.

ANSWER: The trials are precise and accurate: they cluster tightly and center on the accepted value.

L5-2. Trials are 8.42, 8.43, and 8.42 mL; accepted volume is 8.00 mL. Classify accuracy and precision.

ANSWER: Precise but not accurate relative to 8.00 mL.

L5-3. A student reads a meniscus from above. Name the error and give a specific repair.

ANSWER: Parallax/line-of-sight error. Move the eye to liquid level and read the bottom of the concave meniscus.

L5-4. A balance reads +0.20 g when empty. Explain why repeating trials without zeroing it does not solve the problem.

ANSWER: The zero offset is systematic; every result is shifted high by about 0.20 g. Zero or calibrate the balance first.

L5-5. Explain why percent error alone cannot establish precision.

ANSWER: Percent error compares with an accepted value; precision requires repeated trials to judge their agreement.

MISCONCEPTION

A low percent error proves the procedure is precise.

REPAIR

Ask for the repeated-trial data. If no spread can be examined, a precision claim is unsupported.



Cumulative Review Key

Answers from all lessons

TEACHER KEY

R1. Rewrite '.075kgs' as a correct SI-style record.

ANSWER: 0.075 kg.

R2. Write 0.0008200 in scientific notation and state its significant figures.

ANSWER: 8.200×10^{-4} ; four significant figures.

R3. Calculate 6.20×4.1 and report the correct significant figures.

ANSWER: 25; the product 25.42 rounds to two significant figures.

R4. Calculate $12.08 + 0.6 + 3.214$ and apply the decimal-place rule.

ANSWER: 15.9.

R5. Convert 2.75 h to seconds using dimensional analysis.

ANSWER: $2.75 \text{ h} \times 60 \text{ min/h} \times 60 \text{ s/min} = 9.90 \times 10^3 \text{ s}$.



Cumulative Review Key

Answers from all lessons

TEACHER KEY

R6. A sample has mass 24.96 g and volume 10.00 mL. Calculate density.

ANSWER: 2.496 g/mL.

R7. A material has density 3.60 g/mL and volume 7.25 mL. Calculate mass.

ANSWER: 26.1 g.

R8. Experimental density is 2.40 g/mL; accepted density is 2.50 g/mL. Calculate percent error.

ANSWER: 4.00%.

R9. Trials 24.9, 25.0, and 25.1 g surround an accepted value of 25.0 g. Evaluate accuracy and precision.

ANSWER: Precise and accurate: trials cluster tightly and center on the accepted value.

R10. A conversion result has the unit kg·mL/g when the target is mL. Diagnose the setup.

ANSWER: An unwanted mass unit did not cancel; reverse or replace the mass conversion factor before calculating.



Unit Test A - Rationale Key

Ten items per page

TEACHER KEY

1. B - The sample mass is 12.40 g.

A quantitative observation includes a measured number and unit: 12.40 g.

2. C - Graduated cylinder

A graduated cylinder is designed for liquid-volume measurement and supports finer readings than a beaker.

3. A - 0.45 g

The record uses a leading zero, a space, and the correct lowercase unit symbol without a plural s.

4. D - 82.5 mL

Multiply liters by the exact factor 1000 mL/L: $0.0825 \text{ L} = 82.5 \text{ mL}$.

5. B - 5.070×10^{-4}

A normalized coefficient lies between 1 and 10, and the recorded trailing zero preserves four significant figures.

6. A - 3

The decimal point shows that both trailing zeros are recorded as significant, giving three significant figures.

7. C - 39.7

Both factors have three significant figures, so 39.68 rounds to 39.7.

8. D - 22.1

The tenths place is least precise, so the sum 22.126 rounds to 22.1.

9. A - 3600 s / 1 h

Hours must cancel, so hours belong in the denominator and seconds in the numerator.

10. C - 25.0 m/s

$90.0 \times 1000 / 3600 = 25.0 \text{ m/s}$, retaining three significant figures.



Unit Test A - Rationale Key

Ten items per page

TEACHER KEY

11. B - 2.85 g/mL

Density is mass divided by volume: $57.0 / 20.0 = 2.85 \text{ g/mL}$.

12. D - 31.6 g

$m = dV = 1.24 \times 25.5 = 31.62 \text{ g}$, which rounds to 31.6 g.

13. C - 3.33%

$|12.4 - 12.0| / 12.0 \times 100 = 3.33\%$.

14. A - The trials are precise.

The tight grouping supports precision. Accuracy requires comparison with an accepted value.

15. D - Accuracy

Accuracy describes agreement with an accepted value.

16. B - Read at eye level at the bottom of the meniscus.

Eye-level viewing at the bottom of a concave meniscus reduces parallax error.

17. A - Precision requires repeated trials.

Precision is agreement among repeated measurements, so a single percent-error value is insufficient.

18. D - Reorient a factor so the unwanted unit cancels.

Uncanceled units reveal an incomplete setup; conversion factors must be oriented by unit cancellation.

19. C - Density is intensive.

Density does not depend on sample amount under the same conditions, so it is intensive.

20. B - 2.85 g/mL

The value includes a justified numerical result and the compound unit required for density.



Unit Test B - Rationale Key

Ten items per page

TEACHER KEY

1. A - The crystal is colorless.

Colorless is a descriptive observation without a measured number.

2. B - Electronic balance

An electronic balance directly measures mass.

3. D - 6.0 mL

Milliliter uses lowercase m, uppercase L, a space after the value, and no plural s.

4. C - 7.4 mg

1000 micrograms = 1 milligram, so 7400 micrograms = 7.4 mg.

5. C - 812,000

A positive exponent of 5 moves the decimal five places right: 812,000.

6. D - 5

The captive zero and both trailing decimal zeros are significant, for five total.

7. A - 18

The divisor has two significant figures, so 18.24 rounds to 18.

8. B - 99.6

100.0 is precise to the tenths place, so 99.63 rounds to 99.6.

9. D - 86.4 h

$3.60 \text{ days} \times 24 \text{ h/day} = 86.4 \text{ h}$; 24 is an exact defined factor.

10. B - 64.8 km/h

$18.0 \times 3600 / 1000 = 64.8 \text{ km/h}$.



Unit Test B - Rationale Key

Ten items per page

TEACHER KEY

11. C - 2.98 g/mL

Density equals mass divided by volume: $44.7 \text{ g} / 15.0 \text{ mL} = 2.98 \text{ g/mL}$.

12. A - 5.96 cm³

$V = m/d = 14.9 / 2.50 = 5.96 \text{ cm}^3$.

13. B - 4.00%

$|24.0 - 25.0| / 25.0 \times 100 = 4.00\%$.

14. D - Accurate on average but not precise

The mean supports accuracy, but the wide spread shows low precision.

15. B - Precise but not accurate

Tight grouping shows precision; consistent displacement from the accepted value shows inaccuracy.

16. A - A systematic offset

A constant zero offset shifts measurements in one direction and is systematic.

17. C - They reveal the spread and support a precision judgment.

Repeated trials show variation and therefore provide evidence about precision.

18. A - It is a defined relationship.

Defined unit relationships are exact rather than measured.

19. D - One additional recorded decimal place

The trailing decimal zero records an additional place of precision; it does not change the magnitude.

20. A - 5.96 cm³

A quantitative result must include its justified value and unit.